



SEMESTER END EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E. - Computer Science and Engineering	Semester	: IV
Course Name	: Design and Analysis of Algorithms	Max. Marks	: 100
Course Code	: CS42(00)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- What is matching, perfect matching, stable matching and best valid partner with respect to Gale-Shapely algorithm? Consider an execution of the G-S algorithm that returns a set of pairs S for the given set of men M and women W . Prove that "The set S is a stable matching". CO1 (08)
 - Consider two players $P1$ and $P2$ and nodes in the following order for competitive facility location problem. [10,24,17,22,20,18,12,9,17]. If $P1$ moves first and then $P2$, does target bound 35 and 40 are achievable by $P2$? CO1 (06)
 - Illustrate how asymptotic bounds are used for analyzing algorithms. CO1 (06)
- Determine the stable matching set S for the following set of men and women using Gale-Shapley algorithm. CO1 (06)

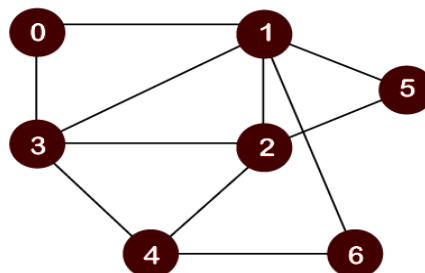
Men's Preference List				
A	R	Q	S	P
B	Q	P	R	S
C	Q	S	P	R
D	R	P	S	Q

Women's Preference List				
P	A	B	D	C
Q	C	A	D	B
R	C	B	D	A
S	B	A	C	D

- Justify the transitivity property of asymptotic growth rate. CO1 (06)
 - Suppose that f and g are two functions such that for some other function h , we have $f = O(h)$ and $g = O(h)$. Then $f + g = O(h)$.
 - If $f = \Omega(g)$ and $g = \Omega(h)$, then $f = \Omega(h)$.
- Justify that every execution of Gale Shapley algorithm for a given set of men M and set of women W results in the set S^* where S^* denote the set of pairs $\{(m, \text{best}(m)), m \in M\}$. CO1 (08)

UNIT - II

- Illustrate the Depth First Search algorithm with an example and find its worst case efficiency when graph is given by adjacency list representation. CO2 (06)
 - Compute Breadth First Search for the following graph. Verify the bipartiteness of the given graph? CO2 (08)

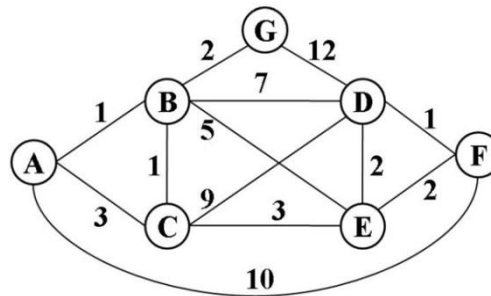


- Prove that if graph $G=(V,E)$ has a topological ordering, then G is a DAG. CO2 (06)

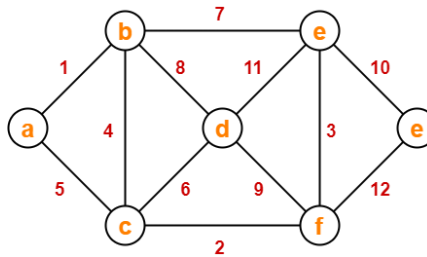
4. a) Write an algorithm to sort the 'n' numbers using Quick sort method. CO2 (06)
 b) Find the number of inversions for the following set of numbers. CO2 (08)
- | | | | | | | | | |
|----|----|----|----|----|----|----|----|-----|
| 56 | 67 | 45 | 78 | 34 | 12 | 98 | 89 | 100 |
|----|----|----|----|----|----|----|----|-----|
- c) Prove that the time complexity of merge sort is $O(n \log n)$ using "unrolling method" and deduce the recurrence relation for the same. CO2 (06)

UNIT- III

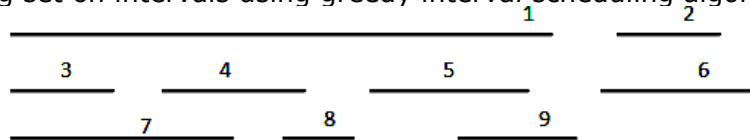
5. a) Construct single source shortest path for the following graph with "A" as the source vertex. Using Dijkstra's algorithm. CO3 (08)



- b) Prove that "Greedy algorithm for interval scheduling returns an optimal set A". CO3 (06)
 c) Construct the minimum spanning tree for the following graph using Prim's algorithm. CO3 (06)



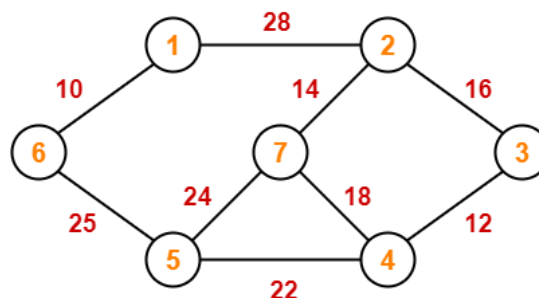
6. a) Identify the optimal set of intervals that can be scheduled for the following set on intervals using greedy interval scheduling algorithm. CO3 (06)



- b) Using Huffman coding, find the average code word length for the following table. CO3 (06)

Character	A	B	C	D	E	F
Frequency	3	5	1	4	2	3

- c) Describe the Kruskal's algorithm to find Minimum Spanning Tree and apply the algorithm for the following graph to find the minimum spanning tree. CO3 (08)

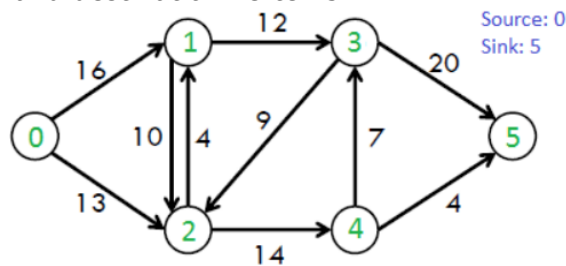


UNIT- IV

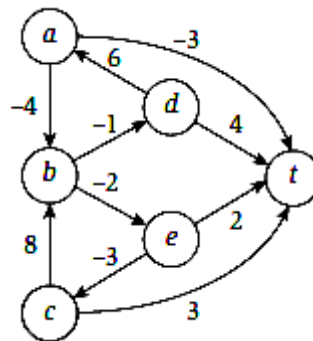
7. a) Discuss survey-design problem for a given set of customers and products CO4 (06)
with an algorithm.
b) Compute the maximum subset-sum that can be achieved for the CO4 (07)
following items with maximum capacity $W=8$.

Item	Weight
1	1
2	2
3	7
4	5

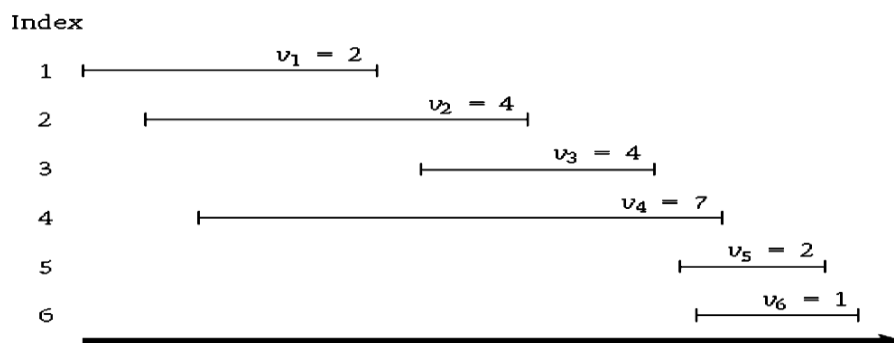
- c) Use ford Fulkerson algorithm to identify the maximum flow between CO4 (07)
source vertex 0 and destination vertex 5.



8. a) Apply bellman ford algorithm to find the distance from source vertex CO4 (08)
'a' to destination vertex 't' in a weighted digraph.



- b) Given the array M of the optimal values of the sub-problems, prove that CO4 (06)
Find_Solution() returns an optimal solution in $O(n)$ time. Illustrate the
proof with this example.

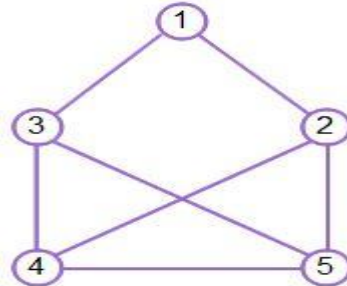


- c) Calculate the maximum value possible by considering the following CO4 (06)
instance and knapsack capacity $W=10$.

Item	Weight	Value
1	4	20
2	7	32
3	5	20
4	3	32

UNIT- V

9. a) What do you mean by polynomial time reduction? Give a correlation between vertex-cover and independent set problem. CO5 (08)
b) Define P, NP, NP-complete and NP-Hard. CO5 (04)
c) Find all Hamiltonian Paths in the given graph. Prove that Hamiltonian Path is NP-Complete. CO6 (08)



10. a) Construct a proof ,Suppose X is an NP-complete problem. Then X is solvable in polynomial time if and only if $P = NP$. CO5 (05)
b) Illustrate with the help of an example, Traveling Salesman Problem in detail. CO5 (07)
c) Explain the general strategy to prove NP-Complete problems. Explain with one example. CO5 (08)
